

BrOSDG



CLARION

Churn Learning with **AI-driven Reasoning and Interpretation**

OCTOBER 2025

Megathon '25

PROBLEM

SIGNIFICANT REVENUE LOSS

Churn requires high acquisition costs to replace lost revenue streams.

LACK OF CLEAR INSIGHT

Traditional methods fail to explain why customers leave, hindering effective retention.

NEED FOR PRECISION

Retention efforts must be targeted and data-driven to maximize ROI.

CHALLENGES

CLASS IMBALANCE

Highly skewed dataset (many negatives/retained, few positives/lost).

INCOMPLETE DATA

Missing key features like claim history (common in public/synthetic datasets).

INTERPRETABILITY

Must understand how to retain customers before they leave.

OUR SOLUTION



Churn Prediction

Develop a robust churn prediction model that neutralizes class imbalance, leveraging the confusion matrix to minimize costly false negatives and maximize the accurate identification of at-risk customers.



Interpretability

Implement SHAP and DeepSHAP for complete model transparency, pinpointing the exact drivers behind every churn prediction to build stakeholder trust.



Interactive Dashboard

Launch an interactive dashboard that demystifies model predictions by providing a clear, visual breakdown of the specific factors driving each customer's churn risk.

DATA PREPROCESSING

Feature Removal: Dropped highly unique or privacy-sensitive identifiers and time-related fields

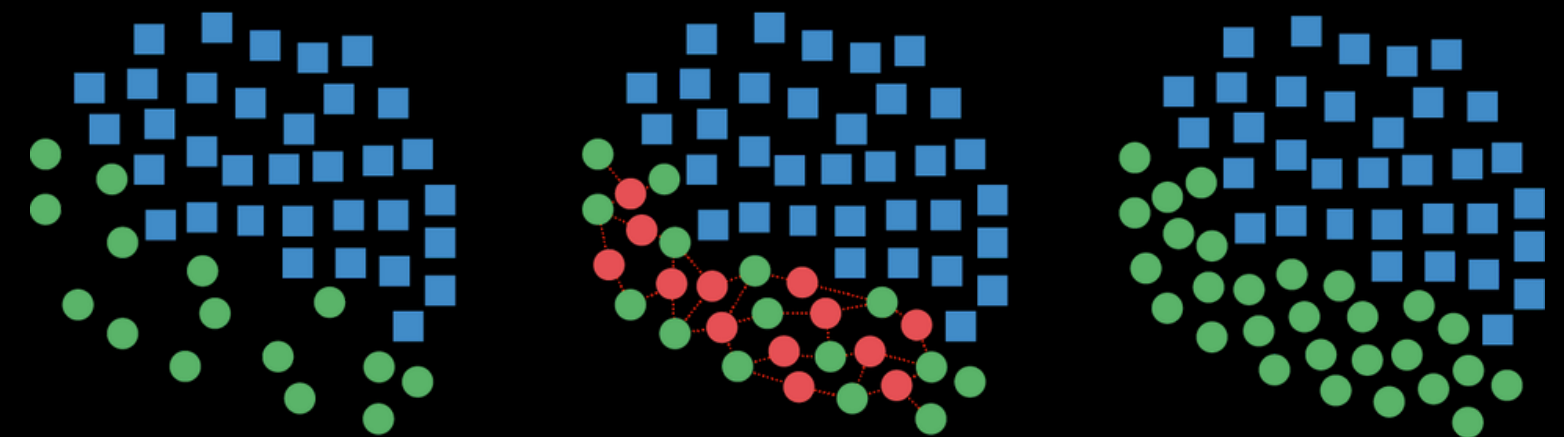
Feature Engineering: Calculated Premium-to-Income Ratio. Replaced Home Market Value ranges with midpoints (0 for non-homeowners) for continuous representation

Data Imputation: Using K-NN for missing Home Market Values

Encoding/Scaling: Target Encoding for high-cardinality features and Z-score Normalisation for numerical features

Data Split: Standard 80% Train, 10% Validation, 10% Test split.

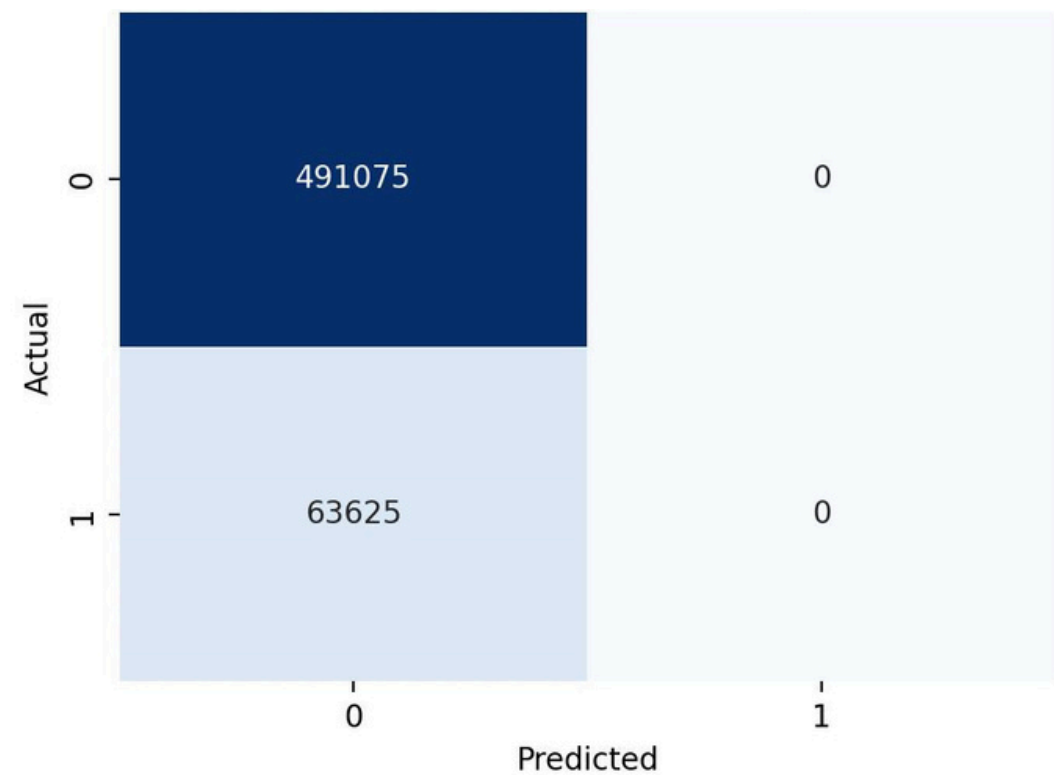
Synthetic Minority Oversampling Technique (SMOTE)



LOGISTIC REGRESSION

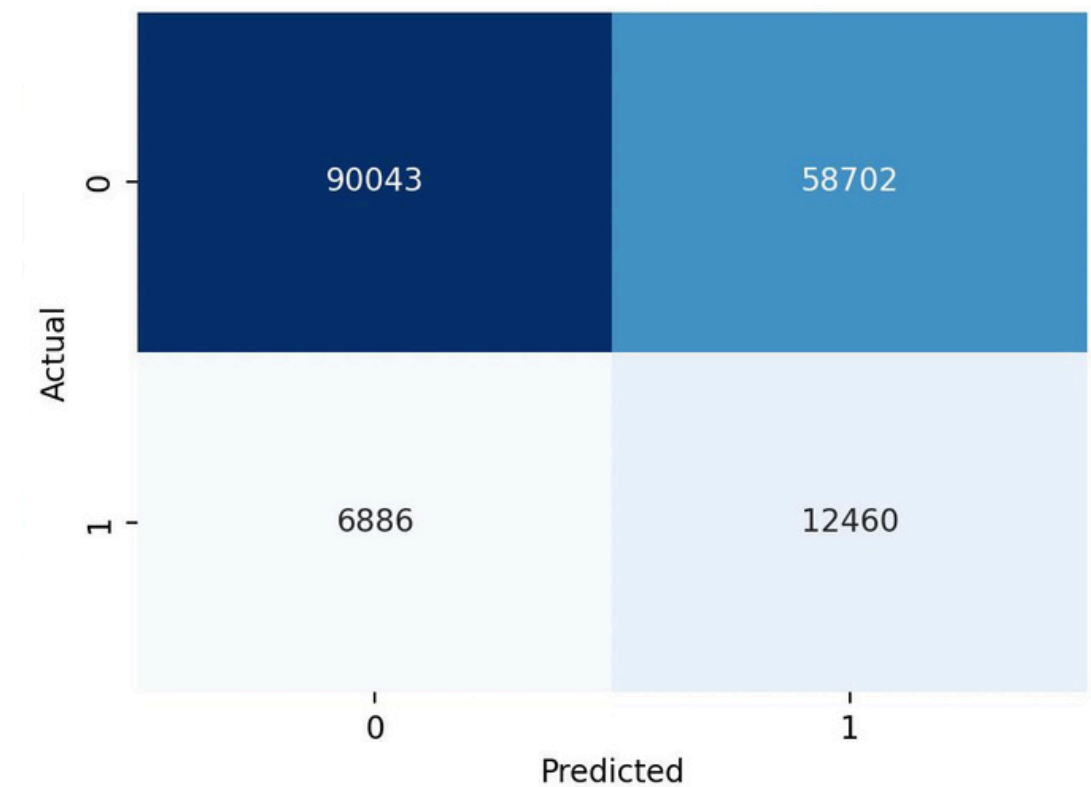
Raw Data Logistic Regression

Due to extreme class imbalance in the raw data, the model learned the simplest solution: always predict the majority class.



Preprocessed Data Logistic Regression

Preprocessing and SMOTE alone made a huge difference!
The dramatic rise in Recall proves the model is now capable of identifying high-risk customers.



MULTI-LAYER PERCEPTRON

The MLP demonstrated a significant jump in overall performance compared to the Logistic Regression baseline, effectively learning the non-linear relationship between features and churn.

Accuracy

0.8575

Precision

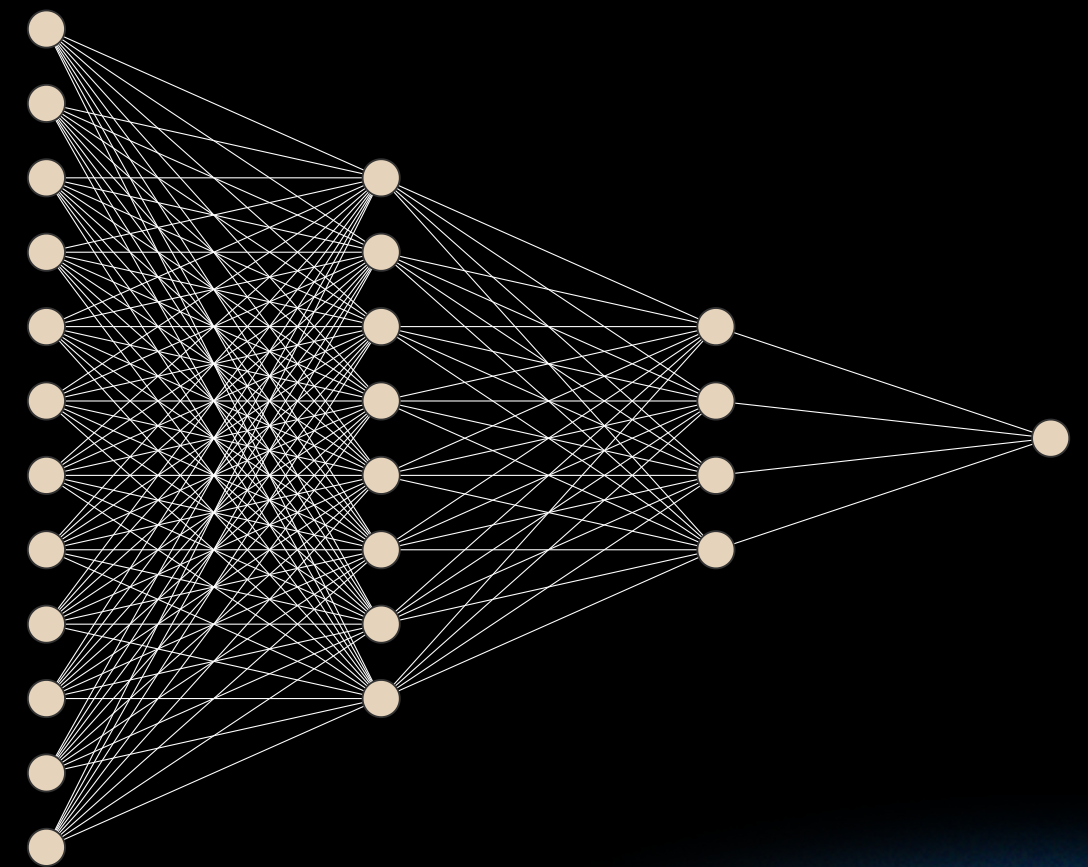
0.3978

Recall

0.4643

F1-score

0.4285



XGBOOST

- Chosen as the industry-standard, high-performance benchmark for tabular data.
- Excellent handling of complex interactions, missing values, and scale.

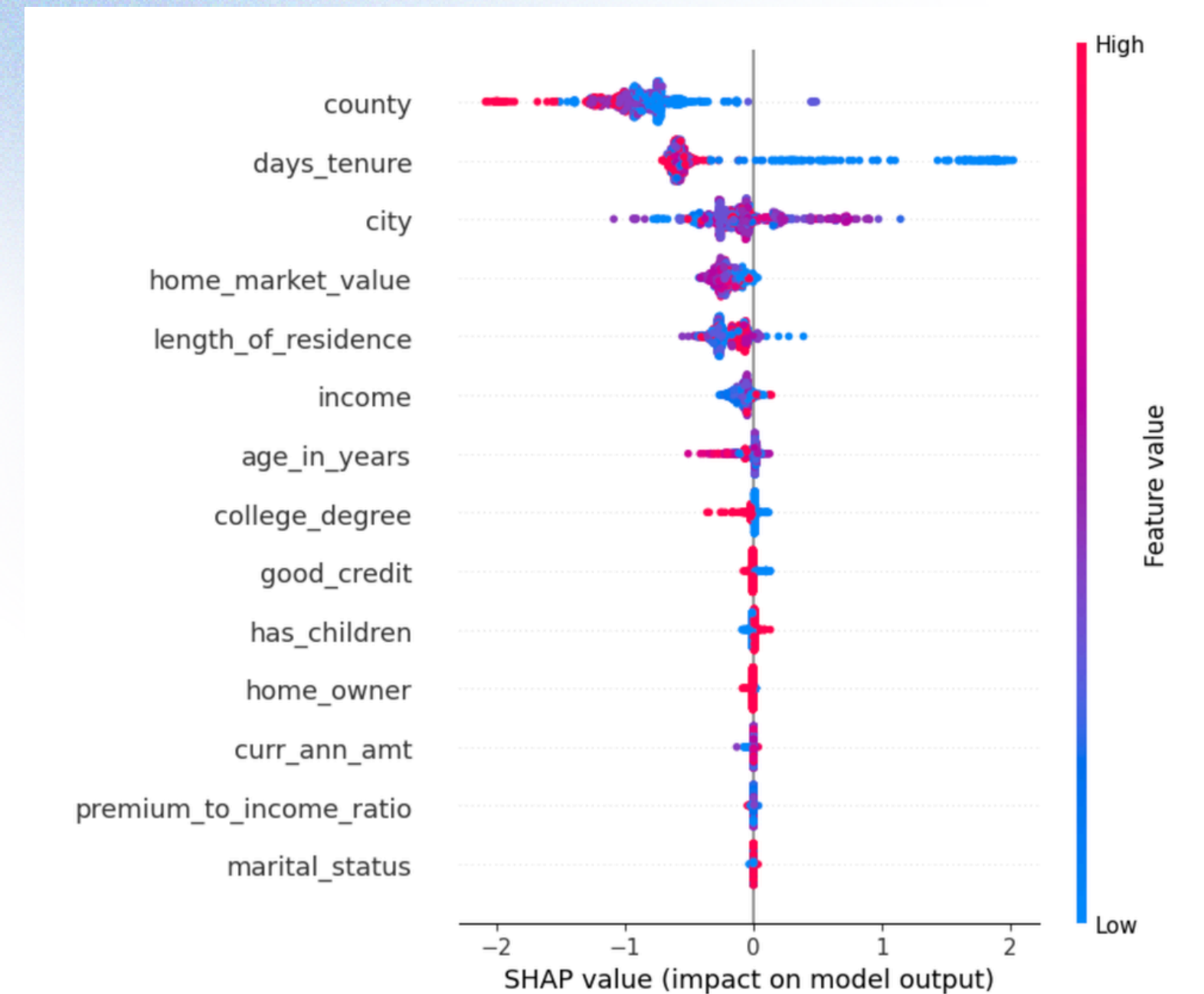


Accuracy 0.88

Precision 0.70

Recall 0.69

F1-score 0.70



TABNET

- Attention-based Deep Learning designed specifically for tabular data
- Provides local, interpretable feature selection through attention masks
- Used DeepSHAP for post-hoc interpretability
- Leverages a two-stage training process

Pre-training: learning to reconstruct masked features in the input rows

Fine-tuning: adapting the learned features to the specific goal of predicting churn

TabNet: Attentive Interpretable Tabular Learning

Sercan Ö. Arık, Tomas Pfister

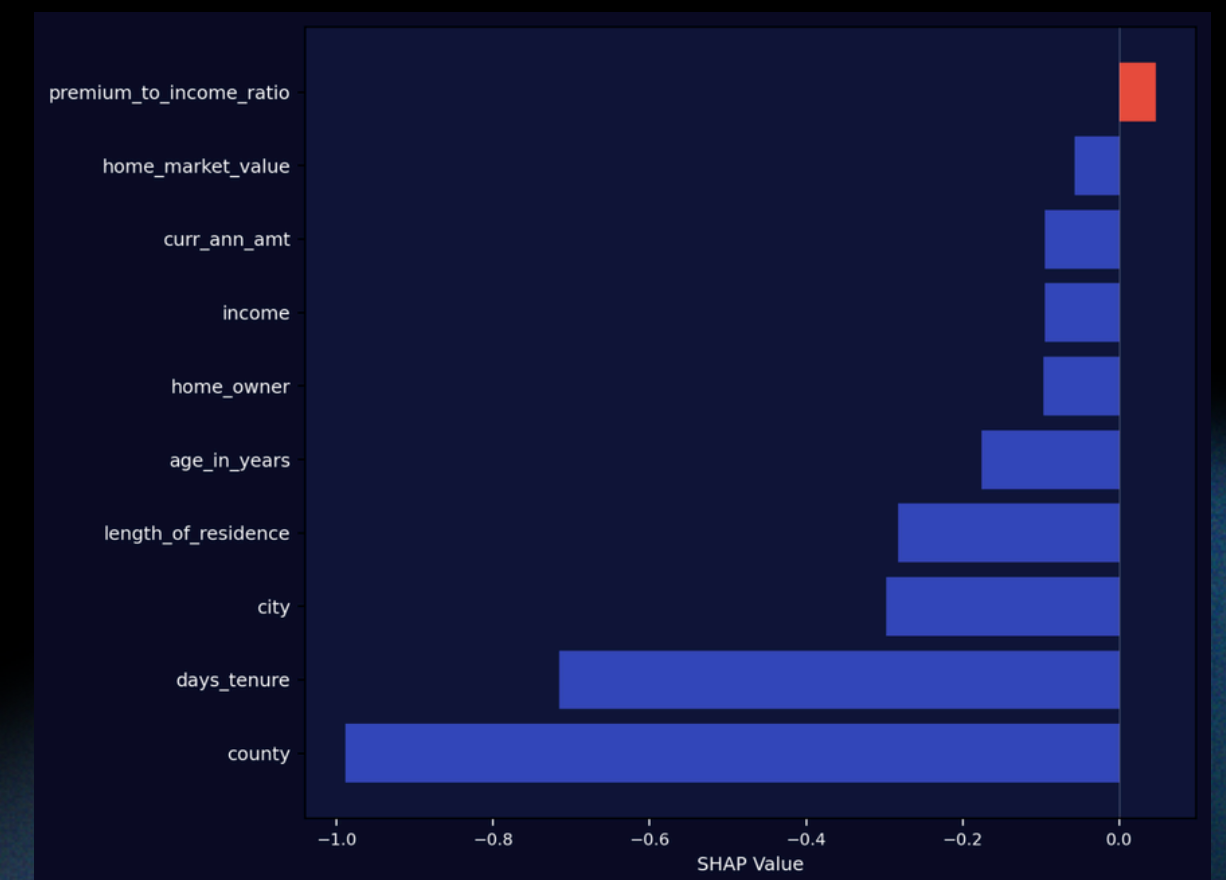
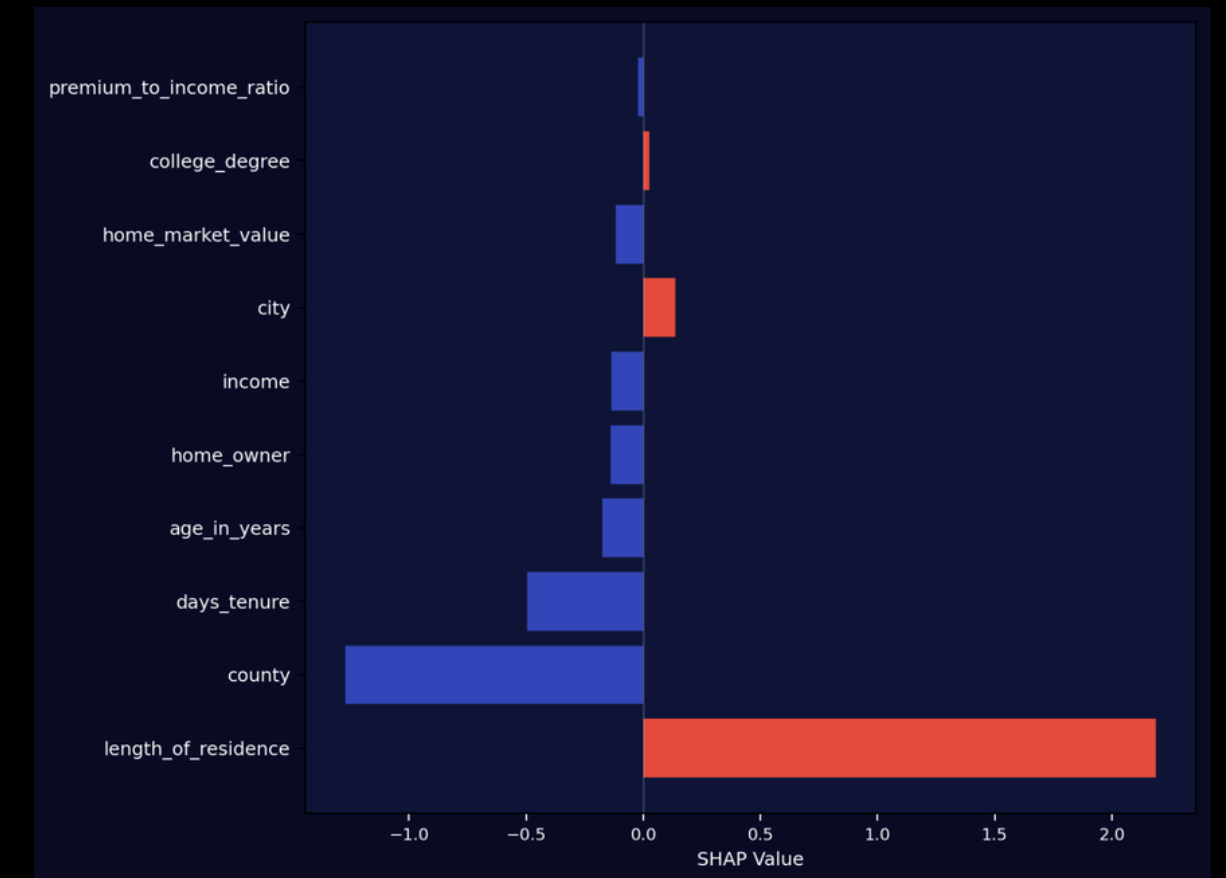
Google Cloud AI

Sunnyvale, CA

soarik@google.com, tpfister@google.com

Accuracy	Precision	Recall	F1-Score
0.85	0.65	0.68	0.66

INTERPRETABILITY (SHAP)



FUTURE DEVELOPMENTS

- **Agent-Based-Insights:** Integrate an agent into the pipeline that can be prompted on plots and a base prompt to output insights and explanations of the visuals. This was not implemented during the hackathon due to the paid API of the agent that allows prompting on images.
- **Downstream Comparator Task:** Develop a comparator to analyze the cost-effectiveness of customer acquisition versus customer retention, specifically focusing on offering incentives to existing customers.
- **A/B Testing for Incentives:** Implement A/B testing to evaluate the effectiveness of different incentive strategies aimed at reducing customer churn.

THANK YOU
